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Studio Session	Activity 1 Thursday 1pm to 3pm

Part A: URL of the Visualisation

<https://public.tableau.com/app/profile/ian.leong4002/viz/GrowingOldinAustralia/GrowingOldinAustralia>

Part B: Domain Description

Describe your domain chosen for DV1

Aged care in Australia. The page follows how population ageing turns into demand for aged care: how many Australians are old, which age groups are growing fastest, whether people are cared for in a residential home or at home, how access differs between the cities and remote areas, what governments spend and how the states and territories compare. Population figures run from 1971 to 2025 with projections to 2045. Aged care figures run from 2017-18 to 2024-25.

Describe the Why and Who aspects of your DV1

Why. One Australian in twelve was 65 or over in 1971. Today it is more than one in six and the 85 and over group, which relies on aged care most, will more than double by 2045. The official figures sit in separate tables from three agencies, so the need is one connected view that shows who uses aged care, how the balance between residential care and care at home has shifted, where access is thinnest and what it costs. The page answers that in a few minutes of reading.

Who. The average Australian, with no statistical training and no knowledge of aged care programmes. That shaped the writing: plain language, no abbreviations, programme names in full, one short paragraph and one headline number per section and a chart title that states the finding, so the reader does not have to work it out.

Describe the “What” aspect of your DV1

Four datasets from three independent sources. Population comes from the Australian Bureau of Statistics, aged care use and spending from the Productivity Commission and residential places and occupancy from the Australian Institute of Health and Welfare with the Department of Health, Disability and Ageing.

Source	Dataset	Used for
Australian Bureau of Statistics	National, state and territory population, December 2025, Table 59. Estimated resident population by age and sex, 1971 to 2025.	Older population by age band, the population pyramid and the denominator of every rate per 1,000.
Australian Bureau of Statistics	Population projections, Australia, 2022 (base) to 2071, Series B, Table B9.	The 2045 projections in the area chart and the pyramid.
Productivity Commission	Report on Government Services 2026, Part F, Section 14, Aged care services.	Recipients per 1,000 people aged 65 and over by programme, state and age group, 2017-18 to 2024-25. Real government spending by programme, 2014-15 to 2024-25.
Australian Institute of Health and Welfare, with the Department of Health, Disability and Ageing	Aged Care Data Snapshot 2025 (GEN Aged Care Data).	Residential places and occupancy by remoteness at 30 June 2025.

Creation process. The spreadsheets were downloaded, reshaped from wide to long form and cut into one tidy file per chart with a Python script. Population and aged care figures are combined only as rates per 1,000 people aged 65 and over, because the Commission counts people over a financial year while the population is a 30 June estimate, so the two cannot be joined row by row. Spending is in constant 2024-25 dollars. Each file is loaded into Tableau as an extract and the full citations with links are printed at the foot of the dashboard.

Describe the “How” aspect of your DV1

The dashboard is one scrolling page of twelve charts in six sections, in the order of the Week 2 sketch. Each section has a heading, a short paragraph, a callout number and two charts side by side. The idiom for each chart was chosen for the question the reader is asking at that point.

Section	Charts	Why these idioms
02 An ageing nation	Stacked area chart of the population aged 65 and over in three age bands. Population pyramid for 2025 and 2045.	The area chart shows the total and the growing share of the 85 and over band at once. The pyramid shows the shape of the age distribution by sex, which is where the thickening at the top is visible.
03 Where people are cared for	Multi-line chart of recipients per 1,000 by programme. Slope chart of 2017-18 against 2024-25.	The lines show the 2021-22 crossover where Home Care Packages overtook residential care. The slope chart strips the series down to the change itself.
04 Who uses aged care	Heatmap of programme by age group. Waffle chart of every 100 people aged 90 and over.	A heatmap compares two categories at once and colour makes the steep rise with age readable before the numbers are read. The waffle turns a percentage into countable people.
05 City and country gap	Sorted bar chart of places per 1,000 by remoteness. Dot plot of occupancy with a line for the national rate.	Bars give an exact ranking of five ordered categories. Dots are used for occupancy because the scale starts at 60 per cent and a bar drawn from 60 would exaggerate the gap.
06 Paying for it	Waterfall chart of the \$19.5 billion increase in spending. Sorted bar chart of 2024-25 spending by programme.	The waterfall shows which programmes added the money, part by part. The bars show where the money goes now.
07 State by state	Dumbbell chart of residential care against Home Care Packages by state. Bump chart ranking the states on Home Care Packages each year.	The dumbbell shows the level and the gap between the two programmes in one row per state. The bump chart is the only idiom that shows South Australia rising from sixth to first while both territories fell to the bottom.

Advanced idioms. Population pyramid, slope chart, heatmap, waffle chart, waterfall chart, dumbbell chart and bump chart, seven in all.

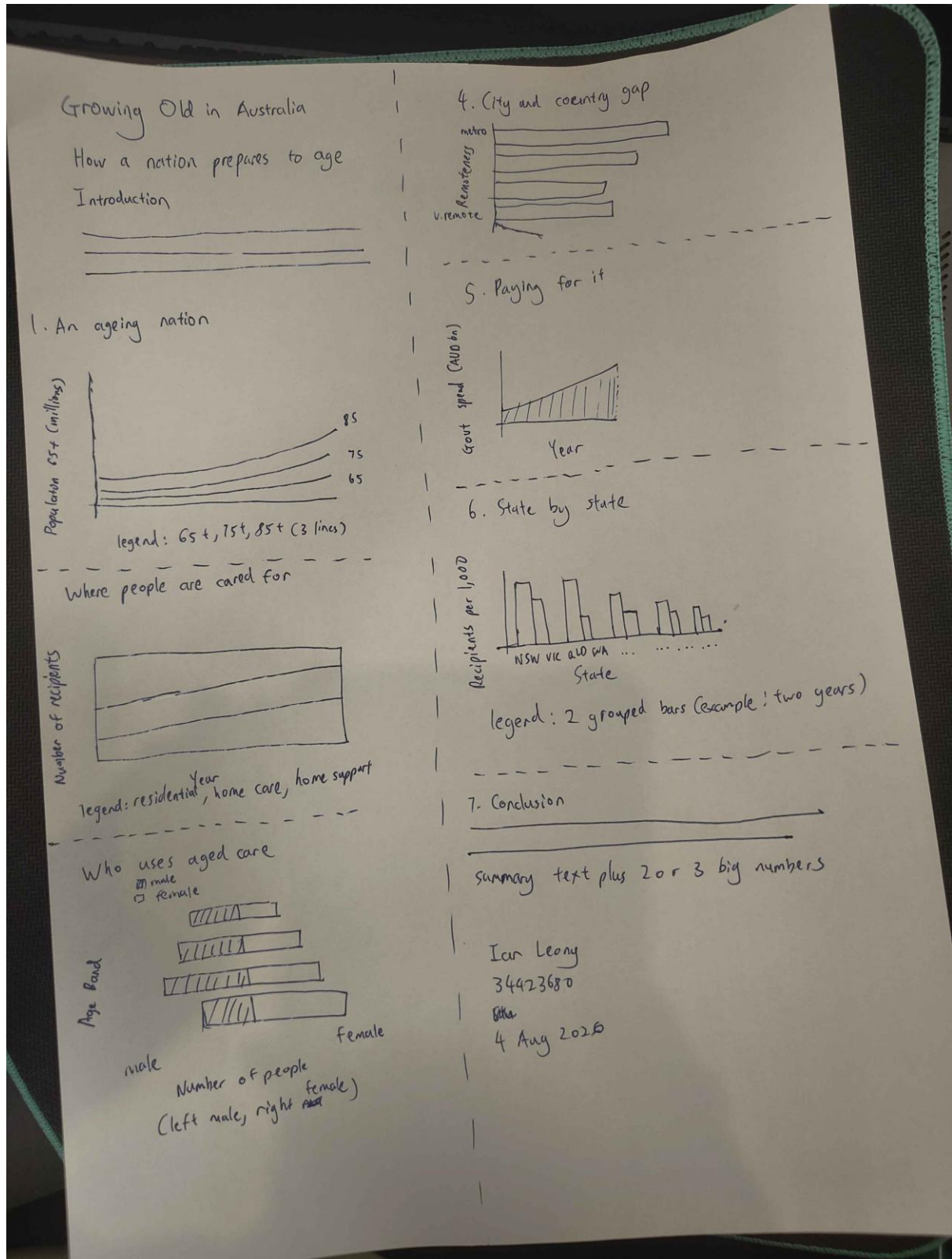
Colour. Blue is residential care, orange is Home Care Packages and green is home support on every chart that shows programmes. Age bands use one blue ramp. The palette is limited to these so colour always means the same thing.

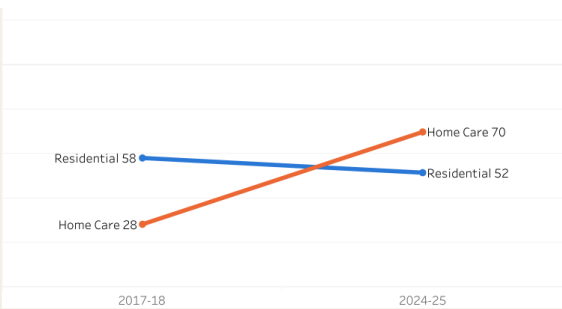
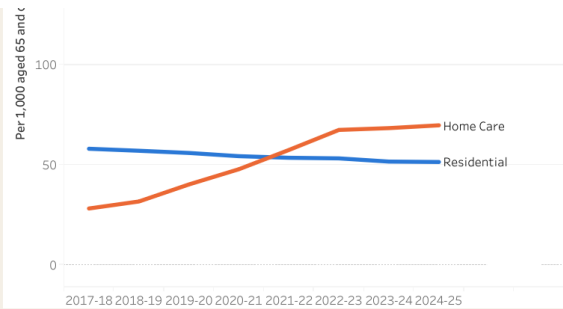
Interaction. Hovering any mark gives a plain-language tooltip with the full year, place and value. Hovering a programme in one chart highlights it in the other charts that show programmes.

Custom-built elements. The bump chart's rank numbers, state names and year labels are placed as text marks on a continuous axis so they never overlap. The dumbbell uses a dual axis. Every chart title sits in its own block of equal height, so paired charts start level. These were built by editing the workbook file, because Tableau's menus do not offer them.

Sketch. The sections follow the Week 2 sketch one for one. The sketch's closing section, a summary with two or three big numbers, became the three tiles beside the introduction and the callout beside each section, so the conclusion is carried by the numbers.

Part C: Screenshot of Week 2 Sketch





04 Who uses aged care

Age, more than anything else, decides whether someone uses aged care. Among Australians aged 65 to 69, four in every thousand spent time in permanent residential care during 2024-25. Among those aged 90 and over, it was 414 in every thousand, more than a hundred times the rate. Put another way: of every 100 Australians aged 90 and over, 41 spent time in permanent residential care last year. Many of the rest were helped at home: 308 in every 1,000 held a Home Care Package and 397 used home support.

414 in 1,000

aged 90 and over used residential care in 2024-25

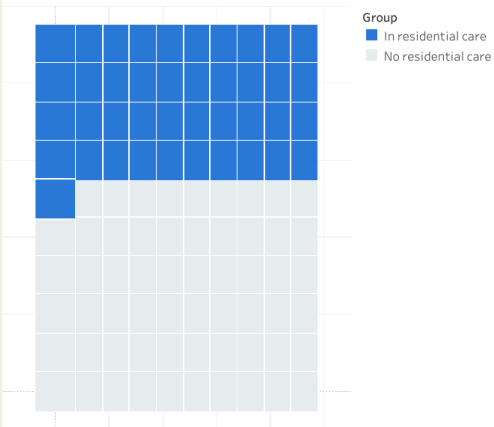
Care use by age and programme, 2024-25

Residential care per 1,000 people in each age group.
4 at ages 65 to 69, 414 at 90 and over.

	65-69	70-74	75-79	80-84	85-89	90 and over
Home Care	9	31	63	124	221	308
Home Support	43	104	190	308	405	397
Residential	4	13	31	73	170	414
Respite	2	5	11	26	56	108
Transition	1	3	5	9	15	18

Of every 100 aged 90 and over, 2024-25

41 spent time in residential care.
The other 59 did not.



05 City and country gap

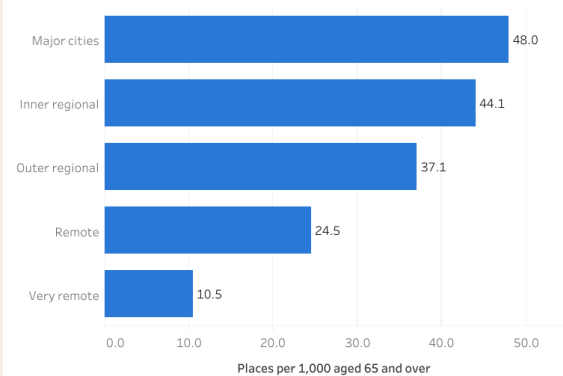
Residential aged care is not spread evenly across the country. For every 1,000 people aged 65 and over, major cities have 48 residential places. Very remote Australia has 10.5, roughly a fifth as many. The places that do exist outside the cities are also the least used: 90 per cent of places were occupied nationally in 2025, but only 76 per cent in very remote areas. Fewer places, and a larger share of them sitting empty. The data shows the pattern, not the cause.

10.5 places

per 1,000 in very remote areas, against 48 in the cities

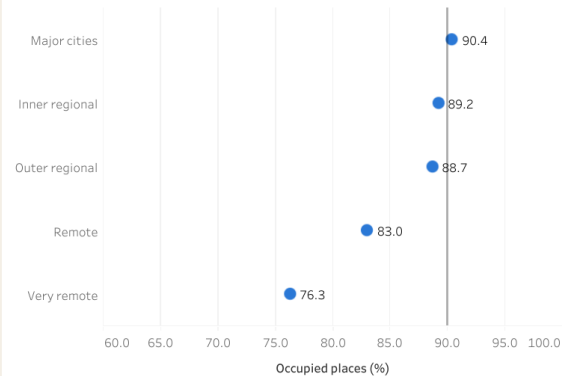
Residential places per 1,000 aged 65 and over, 2025

48 places per 1,000 in the major cities,
10.5 in very remote areas.



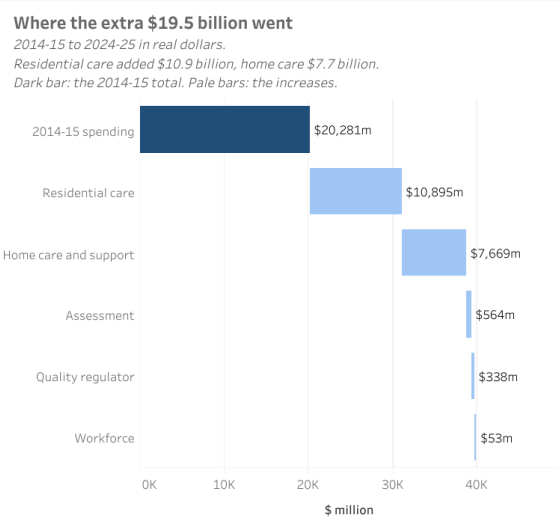
Occupancy of residential places, 2025

Grey line is the national rate, 90%.
Very remote areas sit 13.7 points below it.



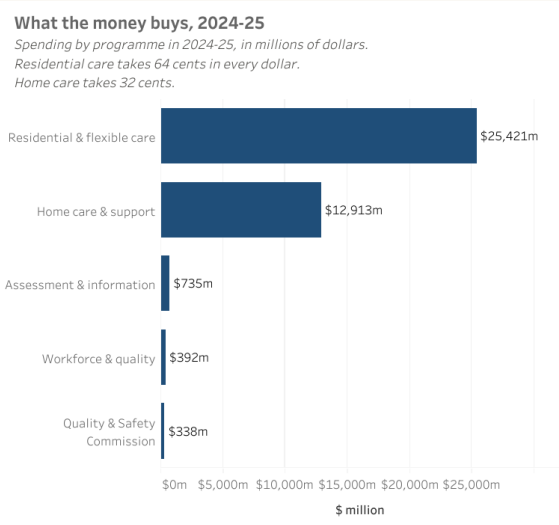
06 Paying for it

Governments spent \$39.8 billion on aged care in 2024-25. A decade earlier, counted in the same dollars, it was \$20.3 billion. Almost all of the \$19.5 billion increase came from two places: residential and flexible care added \$10.9 billion, and home care and support added \$7.7 billion. Home care grew far faster in proportion, to two and a half times its 2014-15 size, yet residential care still takes 64 cents in every dollar.



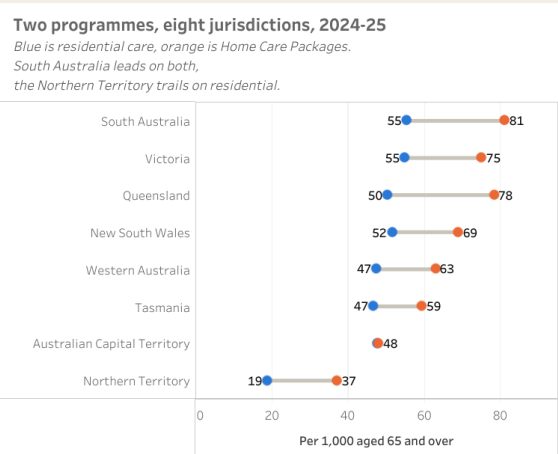
\$19.5 billion

added to aged care spending in a decade, in today's dollars



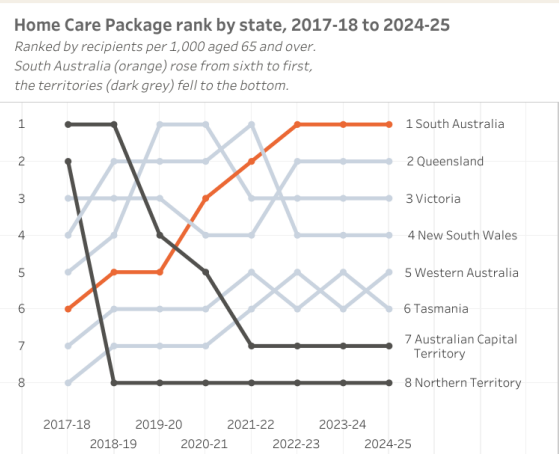
07 State by state

Every state and territory leans on a different mix. South Australia has the highest rate of both permanent residential care (55 per 1,000 aged 65 and over) and Home Care Packages (81). The Northern Territory sits far below every other jurisdiction on residential care, at 19 per 1,000, but it hosts 21 of Australia's 47 National Aboriginal and Torres Strait Islander Flexible Aged Care services, which are counted outside these figures. Its low number is not a simple shortage.



81 in 1,000

South Australians aged 65 and over hold a Home Care Package



Data sources

Productivity Commission. (2026). *Report on Government Services 2026: Part F, Section 14, Aged care services* [Data set].
<https://www.pc.gov.au/ongoing/report-on-government-services/community-services/aged-care-services>
Australian Bureau of Statistics. (2026). *National, state and territory population, December 2025* (Table 59) [Data set].
<https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population>
Australian Bureau of Statistics. (2023). *Population projections, Australia, 2022 (base) to 2071* (Series B, Table B9) [Data set].
<https://www.abs.gov.au/statistics/people/population/population-projections-australia>
Australian Institute of Health and Welfare, & Department of Health, Disability and Ageing. (2025). *Aged care data snapshot 2025* [Data set]. GEN Aged Care Data.
<https://www.gen-agedcaredata.gov.au>